

The switching conjecture for main eigenvalues holds for trees of diameter at most 4

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Abstract

An eigenvalue of a signed graph is *main* if its eigenspace is not orthogonal to the all-ones vector. Akbari, França, Ghasemian, Javarsineh and de Lima conjectured that every connected graph other than K_2 and $K_4 - e$ has a switching, a choice of a sign ± 1 at each vertex, for which all eigenvalues of the resulting signed graph are main, and Akbari, Kumar, Mohar and Pragada asked for a proof for trees and for connected regular graphs. We prove the conjecture for every tree of diameter at most 4 other than K_2 ; this class contains the stars, the double stars and the harmonic trees, for which it was known. Apart from a few explicit ones, the eigenvalues of such a tree are the square roots of the roots of a secular equation. If all neighbours of the centre receive the same sign, each of these whose minimal polynomial is even is main; along two explicit families of switchings, each remaining pair of Galois orbits $\pm O$ spoils at most four members, and at most two when its eigenvalues are irrational. This settles every tree in which a neighbour of the centre has at least 13 leaves; the other cases reduce to explicit switchings and to a finite set of 13,376 trees, which three independently written programs check. Some abstract steps of the proof are formally verified in Lean 4. The results were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

Background. Let G be a finite simple graph with adjacency matrix $A = A(G)$. A *switching* of G is a vector $\mathbf{s} \in \{\pm 1\}^{V(G)}$. The signed graph $G^{\mathbf{s}}$ has the edges of G , the edge uv carrying the sign $s_u s_v$, so that $A(G^{\mathbf{s}}) = D_{\mathbf{s}} A D_{\mathbf{s}}$, where $D_{\mathbf{s}}$ is the diagonal matrix with diagonal $\mathbf{s}$ [2]. An eigenvalue of a signed graph is *main* if its eigenspace is not orthogonal to the all-ones vector $\mathbf{1}$ [3, 11]. A regular graph has only one main eigenvalue, whereas almost all graphs have only main eigenvalues [10]. Akbari, França, Ghasemian, Javarsineh and de Lima [1] asked whether a suitable switching always makes all eigenvalues main, and conjectured that it does. We state the conjecture in the form of [2, Conjecture 1.1].

Conjecture ([1]). *Let G be a connected graph other than K_2 and $K_4 - e$. Then some switching $\mathbf{s}$ of G makes every eigenvalue of $G^{\mathbf{s}}$ main.*

Only distinct eigenvalues matter [2, Proposition 1.3]. The abstract of [1] states the conjecture for every graph other than K_2 and $K_4 - e$; Akbari, Kumar, Mohar and Pragada [2] state it for connected graphs and show that connectedness is needed (no switching of $K_2 \cup K_1$ works). For trees the two versions agree.

According to its abstract, [1] proves the conjecture for Cayley, distance-regular, vertex- and edge-transitive graphs, double stars and paths. It also covers complete bipartite graphs, hence

stars ([1, Theorem 4.1], as cited in [12, Section 4]), and, by a computer search, all connected graphs on at most nine vertices other than K_2 and $K_4 - e$ (as reported in [2, Section 1.2] and [12, Section 5]). Shao and Yuan [12] proved it for harmonic trees [12, Proposition 2.1], complete multipartite graphs and the graphs $S_{n,r}$, and Stanić [13] gave sufficient conditions for net-regular signed graphs. The authors of [2] proved two asymptotic versions, with $n - O(n/(\log n)^{1/4})$ main eigenvalues counted with multiplicity and with $d - O(d/(\log d)^{1/4})$ distinct main eigenvalues, where n is the order and d the number of distinct eigenvalues. They single out regular graphs and trees as unresolved cases and, in [2, Problem 1.6], ask to show that the conjecture “holds for trees and connected regular graphs of order $n \geq 3$ ”.

Result.

Theorem 1. *Let T be a tree of diameter at most 4 other than K_2 . Then some switching $\mathbf{s}$ of T makes every eigenvalue of $T^{\mathbf{s}}$ main.*

The trees of diameter at most 4 are the trees of radius at most 2. They include the stars, the double stars (all trees of diameter 3), the paths P_3, P_4, P_5 , and the harmonic trees of [12], in which a vertex of degree $a^2 - a + 1$ has only neighbours of degree a and all other vertices are leaves. So Theorem 1 contains the known cases above of diameter at most 4 and adds all other trees of diameter 4. Since every signature of a tree is a switching of the all-positive one, Theorem 1 also gives the signature version of the conjecture [2, Conjecture 1.7] for these trees.

Method. A tree of diameter at most 4 with at least three vertices is a rooted tree $T(\mathbf{a})$ of height at most 2: a centre c with neighbours $v_1, \dots, v_k$, where v_i has a_i further neighbours, all of them leaves. Apart from 0 and the numbers $\pm\sqrt{b}$ for branch sizes $b \geq 1$ that occur more than once, its eigenvalues are the numbers $\pm\sqrt{t}$ for the roots t of a *secular polynomial* R (Lemma 3.1). Whether an eigenvalue becomes main after a switching is a simple condition (Lemma 3.2), which for the secular eigenvalues depends only on the Galois orbit. If all neighbours of the centre receive the same sign, every such orbit whose minimal polynomial is even is main (Lemma 3.4); the other orbits form *non-even pairs* $\pm O$, with $\theta \in \mathbb{Q}(\theta^2)$ for $\theta \in O$, which are *integer* pairs if $O \subseteq \mathbb{Z}$ and *irrational* pairs otherwise. Along two explicit families of switchings, the *rows* of Section 4, an integer pair spoils at most four members and an irrational pair at most two, by a linear-independence argument; so a row that is long compared with the number of non-even pairs contains a good switching (Corollary 4.4). Integer pairs are rare, because they come from perfect squares that are roots of R , and all roots of R but one are smaller than the largest branch size $b^* = \max_i a_i$ (Lemma 5.2). For $b^* \geq 13$ this makes the row of the largest branches long enough (Proposition 5.3). For $2 \leq b^* \leq 12$ the trees not covered by the rows lie in an explicit finite set of 13,376 trees, and a computer search leaves three trees with at most seven vertices, for which we give switchings (Propositions 5.5 and 5.6). Trees with $b^* \leq 1$ are treated by explicit switchings (Proposition 5.1).

Scope. One step of the proof, Proposition 5.5, is a finite computation, carried out by three independently written programs (Section 6); all other steps are proved by hand, and some abstract steps are also formally verified (Section 7). The problem remains open for trees in general and for regular graphs; our argument relies on the explicit spectrum of Lemma 3.1. As supporting evidence, the conclusion of Theorem 1 holds for all 63,242,254 trees with 3 to 24 vertices, and direct certificates confirm it for every tree of diameter at most 4 with at most 44 vertices (Section 6).

Prior work. The full text of [1] was not accessible to us: the publisher’s page refused automated access, and we found no other copy. Our account of its results rests on its abstract,

as shown by zbMATH Open, and on the summaries in [12] and [2]. The problem appeared in version 1 of [2], submitted to the arXiv on 22 September 2026. We searched for earlier or concurrent work on the conjecture for trees, in particular for trees of diameter at most 4, on 26 September 2026 between 18:48 and 19:49 UTC and between 21:38 and 22:01 UTC, briefly again at 23:00 UTC, and on 27 September 2026 between 00:37 and 00:42 UTC; the last search was made by the referee agent described in the disclosure. The sources were: the arXiv (the listing of [2], which had only version 1, the arXiv API, and the new submissions in math.CO); zbMATH Open, Semantic Scholar, OpenAlex and Crossref, including the works citing [1] and [12]; the problem database MathDB (<https://mathdb.com>), whose entry for the conjecture was still open, with no solutions; MathOverflow and Mathematics Stack Exchange; GitHub code, commits, issues, pull requests and repositories; public repositories of AI-assisted work on open problems, among them `trureturing` [14]; the publication lists of two authors of [1]; and web search. We did not search Google Scholar or MathSciNet, and dblp and some repository search engines could not be queried, because they returned checks for automated access, which we did not bypass. We found no proof or claim of the conjecture for all trees of diameter at most 4, or for a class of trees containing them; the tree classes treated in the work we found are those listed above. According to its abstract, [5] determines the unsigned trees of diameter 4 with exactly three main eigenvalues; it does not concern switchings. A final check on 27 September 2026 between 02:04 and 02:06 UTC (the arXiv, where [2] still had only version 1; GitHub; MathDB; and MathOverflow) found nothing new. This reports the coverage of our searches, not a guarantee of priority.

2 Switchings and main eigenvalues

Let A be a real symmetric $n \times n$ matrix. For an eigenvalue θ of A let E_θ be its eigenspace and P_θ the orthogonal projection onto E_θ . For $\mathbf{s} \in \{\pm 1\}^n$ the matrix $D_{\mathbf{s}}AD_{\mathbf{s}}$ has the same eigenvalues, with eigenspaces $D_{\mathbf{s}}E_\theta$ and projections $D_{\mathbf{s}}P_\theta D_{\mathbf{s}}$.

Lemma 2.1. *An eigenvalue θ of A is a main eigenvalue of $D_{\mathbf{s}}AD_{\mathbf{s}}$ if and only if $P_\theta \mathbf{s} \neq 0$.*

Proof. The eigenspace $D_{\mathbf{s}}E_\theta$ is orthogonal to $\mathbf{1}$ if and only if the projection $D_{\mathbf{s}}P_\theta D_{\mathbf{s}}\mathbf{1} = D_{\mathbf{s}}P_\theta \mathbf{s}$ of $\mathbf{1}$ onto it vanishes. $\square$

We call $\mathbf{s}$ *good* for A , or for a graph G with $A = A(G)$, if $P_\theta \mathbf{s} \neq 0$ for every eigenvalue θ of A . By Lemma 2.1, Theorem 1 says that every tree of diameter at most 4 other than K_2 has a good switching. If $\mathbf{s}$ is good, so is $-\mathbf{s}$.

Lemma 2.2 (Krylov certificate). *Let A be a symmetric integer matrix with d distinct eigenvalues, let $\mathbf{s} \in \mathbb{Z}^n$, and let K be the integer matrix with columns $\mathbf{s}, A\mathbf{s}, \dots, A^{d-1}\mathbf{s}$. Then $\text{rank}_{\mathbb{Q}} K$ is the number of eigenvalues θ with $P_\theta \mathbf{s} \neq 0$. In particular, $\mathbf{s}$ is good if K has rank d over $\mathbb{F}_p$ for some prime p .*

Proof. We have $A^j \mathbf{s} = \sum_{\theta} \theta^j P_\theta \mathbf{s}$, the nonzero vectors $P_\theta \mathbf{s}$ are orthogonal, and the Vandermonde matrix $(\theta^j)_{\theta, 0 \leq j < d}$ is invertible. So the column space of K is spanned by the vectors $P_\theta \mathbf{s}$, and its dimension is the number of nonzero ones. Reduction modulo p does not increase the rank. $\square$

3 The spectrum of a tree of diameter at most 4

Notation. Let $\mathbf{a} = (a_1, \dots, a_k)$, with $k \geq 1$, be a multiset of nonnegative integers. The tree $T(\mathbf{a})$ has a vertex c , the *centre*, adjacent to $v_1, \dots, v_k$, and v_i is adjacent to a_i further vertices, the leaves in the set L_i . So $T(\mathbf{a})$ has $n = 1 + k + \sum_i a_i$ vertices. We call v_i together with L_i the i th *branch*; it is *bare* if $a_i = 0$. Let B be the set of distinct values of the a_i , let $r = |B|$ and

$b^* = \max B$, and for $b \in B$ let $I_b = \{i : a_i = b\}$ and $k_b = |I_b|$; we put $k_0 = 0$ if $0 \notin B$. We write $B = \{b_1 < \dots < b_r\}$.

A tree with $n \geq 3$ vertices and diameter at most 4 has a vertex of eccentricity at most 2; rooting the tree at such a vertex shows that it is some $T(\mathbf{a})$. Conversely, every $T(\mathbf{a})$ has diameter at most 4. The results of this section and the next hold for every multiset $\mathbf{a}$, so the choice of representation does not matter; a double star, for instance, has two.

Let

$$F(t) = \sum_{b \in B} \frac{k_b}{t - b}, \quad R(t) = \prod_{b \in B} (t - b) - \sum_{b \in B} k_b \prod_{b' \in B \setminus \{b\}} (t - b') = \prod_{b \in B} (t - b) (1 - F(t)).$$

Then $R \in \mathbb{Z}[t]$ is monic of degree r ; we call it the *secular polynomial* of $T(\mathbf{a})$.

Lemma 3.1 (spectrum). *Let $T = T(\mathbf{a})$ and $A = A(T)$.*

- (i) *R has r simple roots $t_1 < \dots < t_r$, with $b_j < t_j < b_{j+1}$ for $1 \leq j < r$ and $t_r > b_r$; in particular $t_j > 0$ and $t_j \notin B$. For each j , both $\theta = \sqrt{t_j}$ and $\theta = -\sqrt{t_j}$ are simple eigenvalues of A , with eigenvector x^θ given by $x_c^\theta = 1$, $x_{v_i}^\theta = \theta/(t_j - a_i)$ and $x_\ell^\theta = 1/(t_j - a_i)$ for $\ell \in L_i$. We call these $2r$ eigenvalues secular.*
- (ii) *For $b \in B$ with $b \geq 1$ and $k_b \geq 2$, both $\theta = \pm\sqrt{b}$ are eigenvalues of multiplicity $k_b - 1$, and E_θ consists of the vectors with $x_{v_i} = \alpha_i$ and $x_\ell = \alpha_i/\theta$ for $\ell \in L_i$ and $i \in I_b$, where $\sum_{i \in I_b} \alpha_i = 0$, and with all other coordinates 0.*
- (iii) *If $k_0 \geq 1$, then $\ker A$ consists of the vectors with $x_c = 0$, with $x_{v_i} = 0$ and $\sum_{\ell \in L_i} x_\ell = 0$ whenever $a_i \geq 1$, and with $\sum_{i \in I_0} x_{v_i} = 0$. If $k_0 = 0$, then $\ker A$ consists of the vectors with $x_{v_i} = 0$ and $\sum_{\ell \in L_i} x_\ell = -x_c$ for every i .*
- (iv) *A has no other eigenvalues. Hence A has $d = 2r + 2|\{b \in B : b \geq 1, k_b \geq 2\}| + [\ker A \neq 0]$ distinct eigenvalues.*

Proof. The equation $Ax = \theta x$ reads

$$\theta x_\ell = x_{v_i} \quad (\ell \in L_i), \quad \theta x_{v_i} = x_c + \sum_{\ell \in L_i} x_\ell, \quad \theta x_c = \sum_{i=1}^k x_{v_i}. \quad (1)$$

Let $\theta \neq 0$ and $x \neq 0$ be a solution. The first two equations give $x_\ell = x_{v_i}/\theta$ and $(\theta^2 - a_i)x_{v_i} = \theta x_c$. If $x_c \neq 0$ we may take $x_c = 1$; then $\theta^2 \neq a_i$ for all i , $x_{v_i} = \theta/(\theta^2 - a_i)$, and the third equation becomes $F(\theta^2) = 1$, that is, $R(\theta^2) = 0$. Conversely, these formulas solve (1) whenever $R(\theta^2) = 0$ and $\theta^2 \notin B$. If $x_c = 0$, then $x_{v_i} = 0$ unless $a_i = \theta^2$, so $\theta^2 = b \in B$ with $b \geq 1$, and the third equation gives $\sum_{i \in I_b} x_{v_i} = 0$; this is (ii), and it needs $k_b \geq 2$. For $\theta = 0$ the equations say $x_{v_i} = 0$ when $a_i \geq 1$, $x_c + \sum_{\ell \in L_i} x_\ell = 0$ for every i , and $\sum_i x_{v_i} = 0$; a bare branch forces $x_c = 0$, and (iii) follows.

The value $R(b_j) = -k_{b_j} \prod_{i \neq j} (b_j - b_i)$ has sign $(-1)^{r-j+1}$, and $R(t) \rightarrow +\infty$ as $t \rightarrow \infty$. So R changes sign on each interval (b_j, b_{j+1}) and on (b_r, ∞) , which gives r roots as stated; there are no others, since $\deg R = r$. As no t_j lies in B , the eigenvalues in (i), (ii) and (iii) are distinct, and those in (i) are simple. Finally the dimensions add up to n : they are $2r$ in (i), $2\sum_{b \geq 1} (k_b - 1)$ in (ii), and $\sum_{a_i \geq 1} (a_i - 1) + k_0 - 1$ if $k_0 \geq 1$, or $1 + \sum_i (a_i - 1)$ if $k_0 = 0$, in (iii); in both cases the total is $1 + k + \sum_i a_i$. $\square$

The effect of a switching. For a switching $\mathbf{s}$ of $T(\mathbf{a})$ we write $s_c = s(c)$, $\sigma_i = s(v_i)$ and $\lambda_i = \sum_{\ell \in L_i} s(\ell)$, the *leaf sum* of branch i ($\lambda_i = 0$ for a bare branch), and for $b \in B$

$$S_b = \sum_{i \in I_b} \sigma_i, \quad \Lambda_b = \sum_{i \in I_b} \lambda_i, \quad A_b = s_c k_b + \Lambda_b.$$

Lemma 3.2 (main eigenvalues after switching). *A switching $\mathbf{s}$ of $T = T(\mathbf{a})$ is good if and only if the following three conditions hold.*

(S) *For every secular eigenvalue θ , with $t = \theta^2$,*

$$G_{\mathbf{s}}(\theta) := \sum_{b \in B} \frac{A_b + S_b \theta}{t - b} \neq 0.$$

(L) *For every $b \in B$ with $b \geq 1$ and $k_b \geq 2$ and each $\theta \in \{\sqrt{b}, -\sqrt{b}\}$, the numbers $\sigma_i + \lambda_i/\theta$ with $i \in I_b$ are not all equal.*

(Z) *If 0 is an eigenvalue: when $k_0 \geq 1$, some branch with $a_i \geq 2$ has leaves of both signs, or the σ_i with $i \in I_0$ are not all equal; when $k_0 = 0$, some branch has leaves of both signs, or $s_c \neq \sum_i \varepsilon_i$, where ε_i is the common sign of the leaves in L_i .*

Proof. By Lemma 3.1, $\mathbf{s}$ is good if and only if $\mathbf{s}$ is not orthogonal to x^θ for each secular θ , nor to $E_{\pm\sqrt{b}}$, nor to $\ker A$. First, $\mathbf{s} \cdot x^\theta = s_c + \sum_i (\sigma_i \theta + \lambda_i)/(t - a_i)$; writing $s_c = s_c F(t)$ and grouping the branches by size gives $G_{\mathbf{s}}(\theta)$. Second, for $x \in E_\theta$ as in Lemma 3.1(ii) we have $\mathbf{s} \cdot x = \sum_{i \in I_b} \alpha_i (\sigma_i + \lambda_i/\theta)$, which vanishes for all α with $\sum \alpha_i = 0$ exactly when the coefficients are all equal. Third, let $k_0 \geq 1$. By Lemma 3.1(iii), $\mathbf{s} \perp \ker A$ if and only if $\mathbf{s}$ is constant on each L_i with $a_i \geq 2$ and on $\{v_i : i \in I_0\}$. Let $k_0 = 0$. If two leaves ℓ, ℓ' of one branch have opposite signs, then $e_\ell - e_{\ell'} \in \ker A$ and $\mathbf{s} \cdot (e_\ell - e_{\ell'}) \neq 0$. Otherwise $\mathbf{s} \cdot x = x_c(s_c - \sum_i \varepsilon_i)$ for $x \in \ker A$, and $\ker A$ contains a vector with $x_c = 1$. $\square$

Orbits. The polynomial $R(x^2) \in \mathbb{Z}[x]$ is monic and even, and by Lemma 3.1(i) its roots are the $2r$ secular eigenvalues, which are distinct. Let f be a monic irreducible factor of $R(x^2)$ in $\mathbb{Q}[x]$; we call the set of its roots an *orbit*. We say that $\mathbf{s}$ *fails at* an orbit O if $G_{\mathbf{s}}(\theta) = 0$ for some $\theta \in O$. A rational number that is a root of R is an integer, since R is monic with integer coefficients.

Lemma 3.3 (orbits). *Let f be as above, θ a root of f , and $t = \theta^2$.*

- (a) *If $f(-x) = f(x)$, then $\theta \notin \mathbb{Q}(t)$. We call the orbit of f even. (As 0 is not a root of $R(x^2)$, the case $f(-x) = -f(x)$ does not occur.)*
- (b) *If $f(-x) \neq f(x)$, then $\theta \in \mathbb{Q}(t)$, and $\pm f(-x)$ is another irreducible factor of $R(x^2)$, whose orbit is the negative of the orbit of f . We call these two orbits a non-even pair. If $\deg f = 1$, then $\theta \in \mathbb{Z}$ and t is a perfect square that is a root of R (an integer pair); otherwise $t \notin \mathbb{Q}$ (an irrational pair).*
- (c) *If an integer root t of R is not a perfect square, then $\{\sqrt{t}, -\sqrt{t}\}$ is an even orbit.*
- (d) *Let N_I and N_{II} be the numbers of integer and of irrational pairs, and N_{ei} the number of roots of R that are even integers but not perfect squares. Then $N_I + 2N_{II} + N_{ei} \leq r$.*
- (e) *For every switching $\mathbf{s}$, if $\mathbf{s}$ fails at an orbit O , then $G_{\mathbf{s}}$ vanishes on all of O .*

Proof. Let $h \in \mathbb{Q}[t]$ be the minimal polynomial of t . Then $h \mid R$ and $f \mid h(x^2)$. If $\theta \notin \mathbb{Q}(t)$, then $\deg f = [\mathbb{Q}(\theta) : \mathbb{Q}] = 2 \deg h$, so $f = h(x^2)$ is even. If $\theta \in \mathbb{Q}(t)$, then $\deg f = \deg h$; if f were even, say $f = f_1(x^2)$, then $f_1(t) = 0$ would give $\deg f = 2 \deg f_1 \geq 2 \deg h$, which is impossible. So f is not even, and $\pm f(-x)$, which divides $R(x^2)$ because $R(x^2)$ is even, is a different irreducible factor. If $\deg f = 1$, then θ is a rational root of the monic integer polynomial $R(x^2)$, hence an integer. If $\deg f \geq 2$, then $\deg h = \deg f \geq 2$ and $t \notin \mathbb{Q}$. This proves (a) and (b), and (c) holds because $x^2 - t$ is irreducible. For (d), the minimal polynomials of the roots of R are distinct irreducible factors of R : each integer pair uses a factor $t - q^2$, each irrational

pair a factor of degree at least 2, and each root counted by N_{ei} a factor $t - m$ with m not a square; their degrees add up to at most $\deg R = r$. For (e), write $G_{\mathbf{s}}(x) = g(x) / \prod_{b \in B} (x^2 - b)$ with $g \in \mathbb{Q}[x]$; the denominator does not vanish at secular eigenvalues, and if $g(\theta) = 0$ for one root θ of f , then $f \mid g$. $\square$

The following reduction is the starting point of the proof.

Lemma 3.4 (equal signs next to the centre). *Suppose that $\sigma_1 = \dots = \sigma_k = \sigma$. Then $G_{\mathbf{s}}(\theta) = \sigma\theta + P_{\mathbf{s}}(t)$ with $P_{\mathbf{s}}(t) = \sum_{b \in B} A_b/(t - b) \in \mathbb{Q}(t)$ for every secular θ . Hence $\mathbf{s}$ fails at no even orbit, and $\mathbf{s}$ fails at a non-even pair, that is, at one of its two orbits, if and only if $P_{\mathbf{s}}(t) \in \{\theta, -\theta\}$, where θ is any root of the pair and $t = \theta^2$.*

Proof. Here $S_b = \sigma k_b$, so $\sum_b S_b \theta/(t - b) = \sigma \theta F(t) = \sigma \theta$. In an even orbit $\theta \notin \mathbb{Q}(t)$, so $\sigma \theta + P_{\mathbf{s}}(t) \neq 0$. The last statement holds because $G_{\mathbf{s}}(\theta) = 0$ means $P_{\mathbf{s}}(t) = -\sigma \theta$ and $G_{\mathbf{s}}(-\theta) = 0$ means $P_{\mathbf{s}}(t) = \sigma \theta$. $\square$

4 Two rows of switchings

In this section we assume $b^* \geq 2$, that is, some branch has at least two leaves.

Standard leaf sums. For every branch i with $a_i \geq 1$ we fix a number λ_i° with $|\lambda_i^\circ| \leq a_i$ and $\lambda_i^\circ \equiv a_i \pmod{2}$, group by group, after ordering each I_b :

- if $b = 1$: $\lambda_i^\circ = -1$, except $\lambda_i^\circ = 1$ for the smallest $i \in I_1$ when $k_1 \geq 2$;
- if $b \geq 3$ is odd: $\lambda_i^\circ = -1$, except that the two smallest $i \in I_b$ get 1 and -3 when $k_b \geq 2$;
- if $b \geq 2$ is even: $\lambda_i^\circ = 0, -2, 0, -2, \dots$ along I_b .

Then (P1) for $b \geq 1$ with $k_b \geq 2$ the λ_i° with $i \in I_b$ are not all equal, and (P2) for $b \geq 2$ the smallest $i \in I_b$ has $|\lambda_i^\circ| < b$, so the leaves of that branch receive both signs whenever their sum is λ_i° . Put $\Lambda_b^\circ = \sum_{i \in I_b} \lambda_i^\circ$ for $b \geq 1$ and

$$U(t) = \sum_{b \in B, b \geq 1} \frac{\Lambda_b^\circ}{t - b}, \quad U_\beta(t) = U(t) - \frac{\Lambda_\beta^\circ}{t - \beta}.$$

Lemma 4.1 (realizable leaf sums). *Let $\beta \geq 1$ and $k \geq 1$, and let $\mathcal{L}_\beta(k)$ be the set of integers Λ for which there are $\lambda_1, \dots, \lambda_k \in \{-\beta, -\beta + 2, \dots, \beta\}$ with $\sum_i \lambda_i = \Lambda$, not all equal if $k \geq 2$, and, if $\beta \geq 2$, with $|\lambda_i| < \beta$ for some i . Then for $\beta \geq 2$, $\mathcal{L}_\beta(k)$ consists of the $\Lambda \equiv k\beta \pmod{2}$ with $|\Lambda| \leq k\beta - 2$, except $\Lambda = 0$ when $\beta = k = 2$; so $|\mathcal{L}_\beta(k)| = k\beta - 1 - [\beta = k = 2]$. For $\beta = 1$, $\mathcal{L}_1(k)$ consists of the $\Lambda \equiv k \pmod{2}$ with $|\Lambda| \leq k - 2$ if $k \geq 2$, and $\mathcal{L}_1(1) = \{\pm 1\}$; so $|\mathcal{L}_1(k)| = k - 1$ for $k \geq 2$ and $|\mathcal{L}_1(1)| = 2$.*

Proof. Every realizable Λ satisfies the stated parity and $|\Lambda| \leq k\beta$, and $|\Lambda| = k\beta$ would force all λ_i to equal $\pm\beta$. For $(\beta, k, \Lambda) = (2, 2, 0)$ the only candidates are $(0, 0)$ and $(2, -2)$, which are excluded. The case $\beta = 1$ is clear. Conversely, let $\beta \geq 2$ and let Λ be as stated. Put $M = (k\beta - \Lambda)/2$, the number of leaves of sign -1 ; then $1 \leq M \leq k\beta - 1$. For $k = 1$ take $\lambda_1 = \Lambda$. For $k \geq 2$ distribute M as evenly as possible, $m_i \in \{\lfloor M/k \rfloor, \lceil M/k \rceil\}$, and put $\lambda_i = \beta - 2m_i$. If $k \nmid M$, the m_i are not all equal and some m_i lies in $[1, \beta - 1]$. If $M = kq$, then $1 \leq q \leq \beta - 1$, and we use $(q + 1, q - 1, q, \dots, q)$ instead; some m_i lies in $[1, \beta - 1]$ unless $k = 2$, $q = 1$ and $\beta = 2$, which is the excluded case. $\square$

We write $\mathcal{L}_\beta = \mathcal{L}_\beta(k_\beta)$ for $\beta \in B$ with $\beta \geq 1$.

Lemma 4.2 (leaf row). *Let $\beta \in B$ with $\beta \geq 1$. For $\varepsilon \in \{\pm 1\}$ and $\Lambda \in \mathcal{L}_\beta$ let $\mathbf{s}_{\varepsilon, \Lambda}$ be a switching with $s_c = \varepsilon$, with $\sigma_i = 1$ for every branch (bare branches included), with leaf sums $\lambda_i = \lambda_i^\circ$ for $a_i \geq 1$, $i \notin I_\beta$, and with leaf sums on I_β as in Lemma 4.1 with total Λ (branch i gets $(a_i - \lambda_i)/2$ leaves of sign -1). These $2|\mathcal{L}_\beta|$ switchings form the leaf row of β . Every member satisfies (L) and (Z), and for every secular θ , with $t = \theta^2$,*

$$G_{\mathbf{s}_{\varepsilon, \Lambda}}(\theta) = \theta + \varepsilon + U_\beta(t) + \frac{\Lambda}{t - \beta}. \quad (2)$$

No member fails at an even orbit. An integer pair makes at most four members fail, at most two for each ε , and an irrational pair makes at most two members fail.

Proof. (L) holds by (P1) for $b \neq \beta$ and by Lemma 4.1 for $b = \beta$, since all $\sigma_i = 1$. For (Z): if $\beta \geq 2$, Lemma 4.1 gives a branch of size β with $|\lambda_i| < \beta$, whose leaves have both signs; if $\beta = 1$, the group $b^* \geq 2$ keeps its standard leaf sums, and (P2) gives such a branch. Since $A_b = \varepsilon k_b + \Lambda_b$ with $\Lambda_0 = 0$, Lemma 3.4 gives $G = \theta + \varepsilon F(t) + \sum_{b \geq 1} \Lambda_b/(t - b)$, which is (2), and no member fails at an even orbit. By Lemma 3.4, $\mathbf{s}_{\varepsilon, \Lambda}$ fails at a non-even pair if and only if $\varepsilon + U_\beta(t) + \Lambda w \in \{\theta, -\theta\}$, where $w = 1/(t - \beta) \neq 0$; for fixed ε the left-hand side is injective in Λ , so at most two Λ fail. If the pair is irrational, then $t \notin \mathbb{Q}$, so $w \notin \mathbb{Q}$ and $1, w$ are linearly independent over $\mathbb{Q}$. The member fails at the orbit of θ exactly when $\varepsilon + \Lambda w = -\theta - U_\beta(t)$, and at the orbit of $-\theta$ exactly when $\varepsilon + \Lambda w = \theta - U_\beta(t)$; each of these equations has at most one solution $(\varepsilon, \Lambda) \in \mathbb{Z}^2$. $\square$

Lemma 4.3 (bare row). *Suppose $k_0 \geq 1$. For $\varepsilon \in \{\pm 1\}$ and $0 \leq m \leq k_0$ let $\mathbf{s}_{\varepsilon, m}$ be a switching with $s_c = \varepsilon$, with leaf sums λ_i° on every branch with $a_i \geq 1$, with $\sigma_i = 1$ on these branches, and with $\sigma_i = -1$ on exactly m of the bare branches and $\sigma_i = 1$ on the others. These $2(k_0 + 1)$ switchings form the bare row. Every member satisfies (L) and (Z), and for every secular θ*

$$G_{\mathbf{s}_{\varepsilon, m}}(\theta) = \varepsilon + U(t) + \theta - \frac{2m}{\theta}. \quad (3)$$

An even orbit makes at most one member fail, and only if its $t = \theta^2$ is an even integer. An integer pair makes at most four members fail, and an irrational pair at most two.

Proof. (L) holds by (P1), and (Z) by (P2) applied to the group b^* . Now $S_0 = k_0 - 2m$ and $S_b = k_b$ for $b \geq 1$, so $\sum_b S_b/(t - b) = F(t) - 2m/t = 1 - 2m/t$, and (3) follows as before. In an even orbit, 1 and θ are linearly independent over $\mathbb{Q}(t)$, so $G = (\varepsilon + U(t)) + \theta(1 - 2m/t)$ vanishes only if $t = 2m$ and $\varepsilon = -U(t)$; this determines the member. Let $(\theta, -\theta)$ be a non-even pair. The member fails at the orbit of θ if and only if $\varepsilon - 2m/\theta = -\theta - U(t)$, and at the orbit of $-\theta$ if and only if $\varepsilon + 2m/\theta = \theta - U(t)$. For fixed ε each equation has at most one solution m , so at most four members fail. If the pair is irrational, then $1/\theta \notin \mathbb{Q}$, so each equation has at most one solution $(\varepsilon, m) \in \mathbb{Z}^2$. $\square$

Corollary 4.4. *Let $b^* \geq 2$. Then $T(\mathbf{a})$ has a good switching if*

(a') $|\mathcal{L}_\beta| \geq 2N_I + N_{II} + 1$ for some $\beta \in B$ with $\beta \geq 1$, or

(b') $k_0 \geq 1$ and $2(k_0 + 1) > 4N_I + 2N_{II} + N_{ei}$.

In case (a') some member of the leaf row of β is good; in case (b') some member of the bare row.

Proof. By Lemma 4.2, at most $4N_I + 2N_{II}$ of the $2|\mathcal{L}_\beta|$ members of the leaf row fail at some orbit, so under (a') some member fails at none. It satisfies (S) because every secular eigenvalue lies in an orbit, and (L) and (Z) by Lemma 4.2; so it is good by Lemma 3.2. Under (b') the same argument applies to the bare row, with at most $4N_I + 2N_{II} + N_{ei}$ failing members by Lemma 4.3 and Lemma 3.3(c). $\square$

The primes match the labels printed by our programs, where (a) and (b) denote an earlier count with four failing members for every non-even pair.

5 Proof of Theorem 1

5.1 No branch with two leaves

Proposition 5.1. *Let $T = T(\mathbf{a})$ have $n \geq 3$ vertices and $b^* \leq 1$, so that $T = T(1^{k_1}, 0^{k_0})$: the centre has k_0 leaf neighbours and k_1 neighbours with one further leaf each. Then T has a good switching, namely:*

- (a) *if $k_1 = 0$ (the star K_{1,k_0} , $k_0 \geq 2$): $s_c = 1$, and -1 on one leaf (on two leaves if $k_0 = 4$) and $+1$ on the other leaves;*
- (b) *if $k_1 = 1$ and $k_0 \geq 2$: $s_c = 1$; -1 on one leaf neighbour of c and $+1$ on the others; $+1$ on the other neighbour v of c and -1 on the leaf of v ;*
- (c) *if $k_1 \geq 2$ and $k_0 \geq 2$: $s_c = 1$; -1 on one leaf neighbour of c and $+1$ on the others; $+1$ on the k_1 other neighbours of c ; -1 on one of their leaves and $+1$ on the others;*
- (d) *if $k_0 = 1$: one of the two switchings $\mathbf{s}^\pm$ with $s_c = \pm 1$, all $\sigma_i = 1$ and leaf sums λ_i° (defined as in Section 4);*
- (e) *if $k_0 = 0$ (a spider with legs of length 2): the switching $\mathbf{s}^+$.*

Proof. In all cases we use Lemma 3.2.

(a) Here $R(t) = t - k_0$, so $t = k_0$, and $G = (k_0 + E\theta)/t = 1 + E/\theta$, where $E \in \{k_0 - 2, 0\}$ is the sum of the signs of the leaves. $G = 0$ would need $E^2 = k_0$, and $(k_0 - 2)^2 = k_0$ only for $k_0 \in \{1, 4\}$, while $E = 0$ for $k_0 = 4$. (L) is void, and (Z) holds because the leaf signs are not all equal.

(b) Here $R(t) = t^2 - (k_0 + 2)t + k_0$ and $G = 1 + (k_0 - 2)/\theta + (\theta - 1)/(t - 1)$. Substituting $k_0 = t(t - 2)/(t - 1)$, which is $R(t) = 0$, gives $G = (\theta^2 - 2)(\theta^2 + \theta - 1)/(\theta(\theta^2 - 1))$. Now $t = 2$ is not a root, as $R(2) = -k_0$. If $\theta^2 + \theta - 1 = 0$, then $t = \theta^2 = 1 - \theta$ satisfies $t^2 - 3t + 1 = 0$, and subtracting this from $R(t) = 0$ gives $(k_0 - 1)(1 - t) = 0$, which is impossible because $k_0 \geq 2$ and $t \neq 1$. (L) is void, and (Z) holds as in (a).

(c) Here $R(t) = h(t) := t^2 - (1 + k_0 + k_1)t + k_0$. With $A_0 = k_0$, $S_0 = k_0 - 2$, $A_1 = 2k_1 - 2$ and $S_1 = k_1$, and using $F(t) = 1$, we get $G = P + \theta Q$ with $P = (t + k_1 - 3)/(t - 1)$ and $Q = (t - 2)/t$. If G vanishes at θ or at $-\theta$, then $P^2 = tQ^2$, that is, $f(t) := t(t + k_1 - 3)^2 - (t - 2)^2(t - 1)^2 = 0$. Since $h(0) = k_0 > 0$, $h(1) = -k_1 < 0$, $h(k_0 + k_1) = -k_1 < 0$ and $h(k_0 + k_1 + 1) = k_0 > 0$, the roots of h lie in $(0, 1)$ and in $(k_0 + k_1, k_0 + k_1 + 1)$; so h has no integer root and is irreducible over $\mathbb{Q}$. Hence $h \mid f$, that is, the remainder $c_1t + c_0$ of f modulo h vanishes, where

$$\begin{aligned} c_1 &= -k_0^3 - 3k_0^2k_1 + 6k_0^2 - 3k_0k_1^2 + 12k_0k_1 - 13k_0 - k_1^3 + 7k_1^2 - 12k_1 + 8, \\ c_0 &= k_0^3 + 2k_0^2k_1 - 6k_0^2 + k_0k_1^2 - 7k_0k_1 + 13k_0 - 4. \end{aligned}$$

As polynomials in k_0 , both have constant leading coefficients ∓ 1 , so a common root forces their resultant to vanish:

$$\text{Res}_{k_0}(c_1, c_0) = (k_1 - 1)(k_1^6 - 11k_1^5 + 60k_1^4 - 148k_1^3 + 224k_1^2 - 192k_1 + 64) = 0.$$

For $k_1 \geq 2$ the first factor is nonzero, and the sextic has no integer root (an integer root would divide 64, and none of $\pm 1, \pm 2, \dots, \pm 64$ is a root). So (S) holds. (L) holds because the leaves of the k_1 branches do not all have the same sign, and (Z) holds as in (a).

(d) Here $R(t) = t^2 - (k_1 + 2)t + 1$ has discriminant $k_1(k_1 + 4)$, strictly between $(k_1 + 1)^2$ and $(k_1 + 2)^2$, so R is irreducible, $N_I = 0$, and there is at most one non-even pair. By Lemma 3.4, $\mathbf{s}^\pm$ fails at no even orbit, and $P_{\mathbf{s}^+}(t) = 1/t + 2[k_1 \geq 2]/(t - 1)$ and $P_{\mathbf{s}^-}(t) = P_{\mathbf{s}^+}(t) - 2F(t) = P_{\mathbf{s}^+}(t) - 2$. If both failed at the pair, then $P_{\mathbf{s}^+}(t)$ and $P_{\mathbf{s}^+}(t) - 2$ would both lie in $\{\theta, -\theta\}$, so $2 \in \{\pm 2\theta\}$

and $t = 1$, which is not a root. (L) holds by (P1). The eigenvalue 0 does not occur, by Lemma 3.1(iii), since $k_0 = 1$ and all $a_i \leq 1$.

(e) Here $R(t) = t - k_1 - 1$ and $P_{\mathbf{s}^+}(t) = 2[k_1 \geq 2]/(t - 1)$. For $k_1 = 1$, $G = \theta \neq 0$. For $k_1 \geq 2$, $G(\mu) = \mu + 2/(\mu^2 - 1)$ for each secular eigenvalue μ , and $G(\mu) = 0$ means $\mu^3 - \mu + 2 = 0$. This covers both $\mu = \theta$ and $\mu = -\theta$, and both are eigenvalues. The polynomial $x^3 - x + 2$ has no rational root, so it is irreducible, and its discriminant is $-104 < 0$, so it has only one real root. But every conjugate of an eigenvalue of a symmetric rational matrix is again an eigenvalue, hence real; so no eigenvalue satisfies $\mu^3 - \mu + 2 = 0$. (This step is formalised; see Section 7. Directly: $\mu^2 = k_1 + 1$ would give $\mu k_1 = -2$, so $\mu^2 = 4/k_1^2 < k_1 + 1$.) (L) holds by (P1). For (Z), each leaf group is a single leaf, and $\sum_i \varepsilon_i = \Lambda_1^\circ \in \{-1, 2 - k_1\}$ differs from $s_c = 1$. $\square$

5.2 Counting non-even pairs

Lemma 5.2. *Let $b^* \geq 1$, let $g = |\{0, 1, \dots, b^*\} \setminus B|$, so that $r = b^* + 1 - g$, and let sq be the number of integers $q \geq 1$ with $q^2 \leq b^* - 1$ and $q^2 \notin B$. Then $\text{sq} \leq \lfloor \sqrt{b^* - 1} \rfloor$, $\text{sq} \leq g$, $N_I \leq \text{sq} + 1$ and $N_I + 2N_{II} + N_{\text{ei}} \leq r$.*

Proof. By Lemma 3.1(i), the roots $t_1, \dots, t_{r-1}$ lie in $(b_1, b^*) \subseteq [0, b^*)$, and no root of R lies in B . An integer pair has $t = q^2$ with $q \geq 1$ (Lemma 3.3(b)); if $t \neq t_r$, then $q^2 \leq b^* - 1$ and $q^2 \notin B$. Hence $N_I \leq \text{sq} + 1$. The squares counted by sq are distinct elements of $\{1, \dots, b^* - 1\} \setminus B$, so $\text{sq} \leq g$; the bound $\text{sq} \leq \lfloor \sqrt{b^* - 1} \rfloor$ is clear, and the last inequality is Lemma 3.3(d). $\square$

5.3 Large branches

Proposition 5.3. *If $b^* \geq 13$, then (a') holds with $\beta = b^*$. Hence some member of the leaf row of b^* is good.*

Proof. With the notation of Lemma 5.2,

$$2(2N_I + N_{II}) = 3N_I + (N_I + 2N_{II}) \leq 3(\text{sq} + 1) + (b^* + 1 - g) \leq 2\text{sq} + b^* + 4 \leq 2\lfloor \sqrt{b^* - 1} \rfloor + b^* + 4.$$

For $b^* \geq 13$ we have $2\lfloor \sqrt{b^* - 1} \rfloor + 7 \leq b^*$: if $b^* \leq 16$, then $\lfloor \sqrt{b^* - 1} \rfloor = 3$; if $b^* \geq 17$, then $q = \lfloor \sqrt{b^* - 1} \rfloor \geq 4$ satisfies $q^2 + 1 \leq b^*$ and $q^2 + 1 - (2q + 7) = (q - 1)^2 - 7 > 0$. So $2(2N_I + N_{II}) \leq 2b^* - 3$, and since the left-hand side is even, $2N_I + N_{II} \leq b^* - 2$. By Lemma 4.1, $|\mathcal{L}_{b^*}| = k_{b^*}b^* - 1 \geq b^* - 1 \geq 2N_I + N_{II} + 1$, which is (a') . The good member of the leaf row is given by Corollary 4.4, that is, by the counts of Lemma 4.2. $\square$

For $b^* = 12$ the same estimate gives only $2N_I + N_{II} \leq 11 = b^* - 1$; for this reason the smaller values of b^* need a search.

5.4 Branches with two to twelve leaves

For a set $B \subseteq \{0, 1, \dots, b^*\}$ containing b^* , with $r = |B|$ and $\text{sq} = \text{sq}(B)$ as in Lemma 5.2, put

$$\nu(B) = \min(\text{sq} + 1, r), \quad M(B) = 2\nu(B) + \lfloor (r - \nu(B))/2 \rfloor.$$

Lemma 5.4. *Let $2 \leq b^* \leq 12$, and suppose that $T(\mathbf{a})$ satisfies neither (a') , for any β , nor (b') . Then $|\mathcal{L}_\beta| \leq M(B)$ for every $\beta \in B$ with $\beta \geq 1$, and $2(k_0 + 1) \leq 3\nu(B) + r$ if $k_0 \geq 1$. Let $\mathcal{R}$ be the set of multisets $\mathbf{a}$ with $2 \leq b^* \leq 12$ that satisfy these two conditions. Then $\mathcal{R}$ is finite; it has 13,376 elements, and each of them has $n \leq 124$.*

Proof. By Lemma 5.2, $N_I \leq \nu(B)$ and $N_{II} \leq \lfloor (r - N_I)/2 \rfloor$. The function $x \mapsto 2x + \lfloor (r - x)/2 \rfloor$ is increasing, so $2N_I + N_{II} \leq M(B)$, and failing (a') gives $|\mathcal{L}_\beta| \leq 2N_I + N_{II} \leq M(B)$. Failing (b') gives $2(k_0 + 1) \leq 4N_I + 2N_{II} + N_{\text{ei}} = 3N_I + (N_I + 2N_{II} + N_{\text{ei}}) \leq 3\nu(B) + r$. There are finitely many sets B , and for each of them the first condition bounds every multiplicity k_b with $b \geq 1$, because $|\mathcal{L}_b(k)| \geq k - 1$ by Lemma 4.1, and the second bounds k_0 . The count of $\mathcal{R}$ and the bound on n were obtained by enumeration (Section 6). $\square$

Proposition 5.5. *Every multiset in $\mathcal{R}$ except $(2, 2)$, $(2, 0, 0)$ and (3) satisfies (a') or (b').*

This is the computer-assisted step of the proof. It was carried out by three independently written programs, which evaluate N_I , N_{II} and N_{ei} exactly; see Section 6.

Proposition 5.6. *The following switchings are good; the three trees have 7, 6 and 5 vertices.*

- (a) $T(2, 2)$: $s_c = \sigma_1 = \sigma_2 = 1$; the leaves of v_1 get $+1, +1$ and those of v_2 get $-1, -1$.
- (b) $T(2, 0, 0) = D(2, 2)$, the double star with two leaves at each of its two central vertices: $s_c = 1$; $+1$ on the two leaf neighbours of c ; -1 on the third neighbour v of c , and $+1, -1$ on the two leaves of v .
- (c) $T(3) = K_{1,4}$, rooted at a leaf c : $s_c = 1$; $+1$ on the centre v of the star; $+1, -1, -1$ on the other three leaves.

Proof. (a) $R(t) = t - 4$, so the secular eigenvalues are ± 2 , and $G = (2 + 2\theta)/2 = 1 + \theta \in \{3, -1\}$. For (L), at $\theta = \pm\sqrt{2}$ the numbers $1 + 2/\theta$ and $1 - 2/\theta$ differ. For (Z), $k_0 = 0$ and the leaf signs are constant on each branch, with $\sum_i \varepsilon_i = 0 \neq 1 = s_c$. (b) $R(t) = (t - 1)(t - 4)$, and $G = (2 + 2\theta)/t + (1 - \theta)/(t - 2)$ takes the values 4, -2 , 1, 1 at $\theta = 1, -1, 2, -2$. (L) is void, and (Z) holds because the leaves of v have both signs. (c) $R(t) = t - 4$ and $G = \theta/(t - 3) = \theta \neq 0$; (L) is void, and (Z) holds because the three leaves adjacent to v have both signs. $\square$

These three trees are also covered by known results: $K_{1,4}$ is a star, $T(2, 0, 0)$ a double star, and $T(2, 2)$ has seven vertices.

5.5 End of the proof

Proof of Theorem 1. The tree K_1 is trivial, since the switching (1) is good, and K_2 is excluded. Let T have $n \geq 3$ vertices; then $T = T(\mathbf{a})$ for some $\mathbf{a}$ (Section 3). If $b^* \leq 1$, Proposition 5.1 applies. If $b^* \geq 13$, Proposition 5.3 applies. If $2 \leq b^* \leq 12$ and (a') or (b') holds, Corollary 4.4 applies. Otherwise $\mathbf{a} \in \mathcal{R}$ by Lemma 5.4, so $\mathbf{a}$ is $(2, 2)$, $(2, 0, 0)$ or (3) by Proposition 5.5, and Proposition 5.6 applies. $\square$

6 Computations

The programs and logs are in version 1.7.0 of [4], directory `certificates/switching/`; those of the referee agent (see the disclosure) begin with `rv_`. They are written in Python with SymPy [8], except `swtrees.c` and `swtrees_print.c` (in C), and all ran on one computer (macOS, arm64).

The finite search. Proposition 5.5 was checked by three separately written programs.

- (1) `final_region.py` enumerates $\mathcal{R}$ set by set, as in Lemma 5.4. For each tree it finds the integer roots of R exactly, by evaluating R at every integer in the intervals of Lemma 3.1(i), the last one being $(b^*, b^* + k]$; this gives N_I , N_{ei} and the number N_{rat} of rational roots. It then uses $N_{II} \leq \lfloor (r - N_{rat})/2 \rfloor$, which is valid because an irrational pair uses two irrational roots and because (a') and (b') only become harder as N_{II} grows. All but ten trees pass (a') or (b') this way; for these ten, $R(x^2)$ is factored over $\mathbb{Q}$, which gives the exact N_{II} , and the three trees of Proposition 5.6 remain. The run takes half a second.
- (2) `final_region_indep.py` enumerates a cruder superset of 7,892,382 multisets, with only the bounds $2N_I + N_{II} \leq \lfloor (2\lfloor \sqrt{b^* - 1} \rfloor + b^* + 4)/2 \rfloor$ and $2(k_0 + 1) \leq 3\lfloor \sqrt{b^* - 1} \rfloor + b^* + 4$, which follow from Lemma 5.2 as in the proofs of Proposition 5.3 and Lemma 5.4, and no

pruning by B . It finds the integer roots of R by a floating-point screen of $F(t) - 1$ at every integer t , confirming each hit in exact rational arithmetic; a true root gives $|F(t) - 1|$ far below the threshold. For the trees that fail this cheap test it bounds N_{II} from above: it factors $R(t)$ over $\mathbb{Q}$ and counts an orbit as even only if its own Rabin test shows that $h(x^2)$ is irreducible modulo a prime not dividing the leading coefficient. It finds the same ten trees and the same three survivors.

- (3) The referee's `rv_region.py` shares no code with (1) and (2). It enumerates $\mathcal{R}$ in another way, over all multiplicity vectors under global caps, with the bounds of Lemma 5.4 computed by brute force over the feasible $(N_{\text{I}}, N_{\text{II}}, N_{\text{ei}})$, and writes the set to `rv_region_list.txt`; a comparison made separately from the program, and repeated when the certificates were packaged, shows that this set equals that of (1). It computes N_{I} , N_{II} and N_{ei} exactly for every tree and certifies each orbit by its own tests: irreducibility by a Rabin test, evenness by a simple root of h modulo a prime that is a quadratic non-residue, and non-evenness by an explicit factorisation $h(x^2) = \pm g(x)g(-x)$. Over $\mathcal{R}$, 13,199 trees have no non-even pair, 176 have one and one tree has two. It finds the same three survivors.

Independently of Sections 3–5, the referee's `rv_certify.py` found for each of the 13,376 trees of $\mathcal{R}$ a switching that is good by Lemma 2.2, with ranks modulo $2^{31} - 1$ and 2147483629 and with d computed exactly, in agreement with Lemma 3.1(iv); for the 3,029 trees with $n \leq 40$ it also computed the rank over $\mathbb{Q}$. Our `final_region_witness.py` checked that for each of the other 13,373 trees of $\mathcal{R}$ the row given by (a') or (b') contains a certified good member and has no more failing members than Lemmas 4.2 and 4.3 allow.

Explicit switchings. `five_refined.py` checked the switchings of Proposition 5.6 by the rank over $\mathbb{Q}$ and, independently, by testing $(\mu/p)(A)s \neq 0$ for every irreducible factor p of the minimal polynomial μ of A , which for rational s is equivalent to goodness. The referee's `rv_exceptional.py` tested all 2^n switchings of the three trees: 72 of 128, 16 of 64 and 12 of 32 are good, including ours. Its `rv_families.py` redid the computation in Proposition 5.1(c) and certified the switchings of Proposition 5.1 for $2 \leq k_0 \leq 40$ and $k_1 \leq 40$, for stars with $k_0 \leq 80$, and for $k_0 \in \{0, 1\}$ with $k_1 \leq 80$.

Checks of the lemmas. These exact checks are not needed for the proof. `test_refined_rows.py` tested all 41,039 leaf and bare rows (380,520 switchings) of the 11,535 multisets with $5 \leq n \leq 27$ and $b^* \geq 2$ against Lemmas 4.2 and 4.3; `test_bounds.py` confirmed Lemma 5.2, (a') at $\beta = b^*$ when $b^* \geq 13$, and that only the trees of Proposition 5.6 fail (a') and (b'), for the 375,820 multisets with $4 \leq n \leq 44$ and $b^* \geq 2$. The referee checked Lemmas 3.1 and 3.2 on 400 trees against exact ranks (`rv_spectrum.py`), and the rows of 69 trees with non-even pairs and of the 9,114 multisets with $b^* \geq 2$ and $n \leq 26$ (`rv_rows.py`, `rv_endtoend.py`). The bounds of Lemma 4.2 are attained, for instance by $T(15)$ (four failing members) and by $T(18, 0, 0)$ (two, from one irrational pair).

Supporting evidence. Two programs checked the trees with $3 \leq n \leq 24$: both of them for $n \leq 22$, and implementation B alone for $n = 23$ and 24. Each of these 63,242,254 trees has a good switching, certified by Lemma 2.2 with d computed exactly, and the counts for each n agree with sequence A000055 of [9]. Implementation B, `swtrees.c`, generates the trees with the algorithm of Wright, Richmond, Odlyzko and McKay [15] (a port of the NetworkX [6] generator), computes d exactly from $\gcd(\varphi, \varphi')$ (found modulo a 61-bit prime and confirmed by exact division over $\mathbb{Z}$) and computes ranks modulo $2^{31} - 1$. Implementation A, `swtrees_A.py`, is independent in every component (canonical augmentation, SymPy for d , exact rank over $\mathbb{Q}$ by fraction-free elimination); it covers $n \leq 21$, and $n = 22$ with trees from the NetworkX generator (`swtrees_A_nx.py`). Separately, `verify_diam4.py` certified a good switching for each of the

376,324 multisets with $3 \leq n \leq 44$, which include every tree of diameter at most 4 with at most 44 vertices, with $p = 2^{61} - 1$ and d from Lemma 3.1; for $n \leq 30$ it also computed d from $\gcd(\varphi, \varphi')$, with no mismatch. The referee’s `rv_certify.py` confirmed the 23,023 multisets with $n \leq 30$ by the rank over $\mathbb{Q}$. Non-even pairs are rare: of the 376,324 multisets, 373,837 have none, 2,442 have one, 44 have two and one has three (`noneven_census.py`).

7 Formal verification

Some abstract steps of the proof are formalised in Lean 4 (version 4.33.1) with Mathlib [7] (commit 0df444a3), in the files `SwitchingThmH.lean`, `SwitchingRowCount.lean`, `SwitchingDiam4Cubic.lean` and `SwitchingWalkProfile.lean` of directory `lean/Research/` in version 1.7.0 of [4]. Each has an audit file (`...Audit.lean`) that restates or prints the main statements; those of the last three files also check small instances. Table 1 lists the main declarations. There K is any field of characteristic 0, and “1 and w are linearly independent over $\mathbb{Q}$ ” is stated as: $a + cw = 0$ with $a, c \in \mathbb{Q}$ implies $a = c = 0$.

Declaration	Statement	Used in
<code>SwitchingThmH.two_sqrt_add_seven_le</code>	for $b \in \mathbb{N}$: $b \geq 13 \Rightarrow 2\lfloor\sqrt{b-1}\rfloor + 7 \leq b$	Proposition 5.3
<code>SwitchingThmH.row_budget</code>	for $b, \text{sq}, g, N_1, N_2 \in \mathbb{N}$: if $b \geq 13$, $\text{sq} \leq \lfloor\sqrt{b-1}\rfloor$, $\text{sq} \leq g$, $N_1 \leq \text{sq} + 1$ and $N_1 + 2N_2 + g \leq b + 1$, then $2N_1 + N_2 + 1 \leq b - 1$	Proposition 5.3
<code>SwitchingThmH.sol_subsingleton</code>	$w, u \in K$, 1 and w linearly independent over $\mathbb{Q} \Rightarrow$ at most one $(\varepsilon, \Lambda) \in \mathbb{Z}^2$ has $\varepsilon + \Lambda w = u$	Lemmas 4.2, 4.3, irrational pairs
<code>SwitchingThmH.indep_of_irrational</code>	$w \in K$ and $w \neq q$ for all $q \in \mathbb{Q} \Rightarrow$ 1 and w linearly independent over $\mathbb{Q}$	the same
<code>SwitchingRowCount.card_bad_le_two</code>	$u, w, \theta \in K$, $w \neq 0 \Rightarrow$ at most two $\Lambda \in \mathbb{Z}$ have $u + \Lambda w \in \{\theta, -\theta\}$ (a bound on <code>Set.ncard</code> ; the set is finite)	Lemmas 4.2, 4.3, integer pairs
<code>SwitchingDiam4.no_eigenvalue_root_cubic</code>	$A \in \mathbb{Q}^{n \times n}$, $A^T = A$, $\theta \in \mathbb{R}$ a root of the characteristic polynomial of $A \Rightarrow \theta^3 - \theta + 2 \neq 0$	Proposition 5.1(e), for θ and $-\theta$
<code>SwitchingWalkProfile.main_after_switching</code>	Remark 8.3; also <code>annihilator</code> , <code>eigen_nonorth</code> , <code>walkRegular_main_after_switching</code>	not used for Theorem 1

Table 1: The main Lean declarations.

The formal statements are abstract. `row_budget` is the arithmetic of Proposition 5.3; its hypotheses come from Lemma 5.2 and are proved on paper. Of the counts in Lemmas 4.2 and 4.3, only two abstract steps are formalised: `card_bad_le_two`, used for a fixed ε after renaming u, w and the target pair, and the linear-independence step `sol_subsingleton` with `indep_of_irrational`. That a member fails at a pair exactly when one of the displayed equations holds, and that the relevant w , namely $1/(t - \beta)$ or $\mp 2/\theta$, is irrational for an irrational pair, is proved on paper only. In particular the refined count itself, that an irrational pair spoils at most two members of a row, is not formalised as a statement about trees. `no_eigenvalue_root_cubic` is exactly the algebraic step of Proposition 5.1(e), applied to θ and to $-\theta$. (`SwitchingRowCount.lean` also proves `card_bad_le_one_of_ne`, which we do not use.) Everything else is proved on paper only, or by computer (Section 6); in particular

Lemmas 2.1, 2.2, 3.1–3.4, 4.1, 5.2 and 5.4, Corollary 4.4, the rest of Propositions 5.1 and 5.3, the finite search of Proposition 5.5 and the checks of Proposition 5.6 are not formalised.

The recorded axiom audits (`#print axioms`) report only `propext`, `Classical.choice` and `Quot.sound` for every declaration in Table 1; the files contain no `sorry`, `admit` or `native_decide` and declare no axioms. Each file and its audit file were first rebuilt by `lake build`, with exit code 0, in a fresh directory that contained only the project configuration, a root file and these two sources, the prebuilt Mathlib packages being linked in; this was done on 26 September 2026 at 20:12, 22:31, 23:28 and 23:53 UTC for `SwitchingWalkProfile`, `SwitchingDiam4Cubic`, `SwitchingRowCount` and `SwitchingThmH`, and these logs are in `certificates/switching/`. After the last of these rebuilds, one line of the module docstring of `SwitchingThmH.lean` was reworded; no Lean code changed. The files as released are covered by the clean replay of the whole repository at version 1.7.0, made on 27 September 2026 between 01:32 and 01:47 UTC in a fresh directory, with the sources and the configuration copied from the release tree and the Mathlib packages of the pinned commit taken from a shared cache (logs `logs/clean-replay-v1.7.0-*` of [4]): `lake build` of all modules completed its 8758 jobs with exit code 0; the eleven audit files of the repository report no axioms other than the three above; `leanchecker` checked the four switching modules, with exit code 0; and the SHA-256 checksums of all 66 source and configuration files, listed in `logs/lean-sources-sha256.txt` and including that of the released `SwitchingThmH.lean`, were confirmed in the replay directory.

8 Further remarks and questions

Remark 8.1 (how many vertices must be switched?). For a tree T with $n \geq 3$ vertices let $\tau(T)$ be the least value of $\min(|U|, n - |U|)$ over the sets U of vertices for which the switching that is -1 exactly on U is good, and let $a(n)$ be the maximum of $\tau(T)$ over the trees with n vertices. The computations of Section 6 give, for $n = 3, 4, \dots, 24$,

$$a(n) = 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, 3, 3, 4.$$

The lower bounds come from explicit trees, checked exactly by the rank over $\mathbb{Q}$ and by the test $(\mu/p)(A)\mathbf{s} \neq 0$ (`lower_bounds.py`, `crosscheck_galois.py`). The upper bounds rest on implementation B, confirmed by implementation A for $n \leq 20$ (`swtrees_A_tiers.py`). At $n = 24$, B found 39 trees without a certified switching of at most three vertices; exact checks (`swtrees_print.c`, `verify_hard24.py`) show that none has one and that each has a good switching with $|U| = 4$; the checks of seven of these trees were completed when the certificates were packaged (`certificates/switching/checks-addition/`). For instance, the tree on the vertices $0, \dots, 12$ with edges $0-1, 1-2, 2-3, 3-4, 0-5, 5-6, 6-7, 5-8, 0-9, 9-10, 9-11$ and $0-12$ needs three switched vertices. So a proof for all trees cannot switch at most three vertices. We do not know whether $a(n)$ is bounded.

Remark 8.2 (pairing in bipartite graphs). Lemma 3.4 reflects a general fact. Let G be bipartite, with $A = \begin{pmatrix} 0 & C \\ C^\top & 0 \end{pmatrix}$, let $\theta \neq 0$ be an eigenvalue whose minimal polynomial is even, and let $\mathbf{s}$ be rational. Then $P_\theta \mathbf{s} = 0$ if and only if $P_{-\theta} \mathbf{s} = 0$. Indeed, $E_{\pm\theta}$ consists of the vectors $(u, \pm C^\top u / \theta)$ with $u \in \ker(CC^\top - \theta^2)$, a space with a basis over $\mathbb{Q}(\theta^2)$, and $\mathbf{s} \cdot (u, \pm C^\top u / \theta) = a \pm \theta b$ with $a, b \in \mathbb{Q}(\theta^2)$, while $\theta \notin \mathbb{Q}(\theta^2)$. So in a tree only the eigenvalues with $\theta \in \mathbb{Q}(\theta^2)$, such as the roots of the factors $x \pm 1$ and $x^2 \pm x - 1$ of the characteristic polynomial of the tree in Remark 8.1, impose separate conditions for θ and $-\theta$.

Remark 8.3 (regular graphs). We do not treat regular graphs, but record a sufficient condition that is formalised in `SwitchingWalkProfile.lean`. Let A be real symmetric of order $n \neq 2$ with constant row sums k , and suppose that $(A^j)_{vv} = \sum_u w_u (A^j)_{uu}$ for all $j \geq 0$, for an index v and weights $w_u > 0$. Then $\mathbf{s} = \mathbf{1} - 2e_v$ is good for A : if $f(A)\mathbf{s} = 0$ for a polynomial f ,

then $f(k)\mathbf{1} = 2Me_v$ for $M = f(A)$, summing coordinates gives $(n-2)f(k) = 0$, so $Me_v = 0$, and $0 = (M^2)_{vv} = \sum_u w_u \|Me_u\|^2$ forces $M = 0$; now take $f = \mu_A/(x - \theta)$. For walk-regular graphs ($w_u = 1/n$) this already follows from Stanić’s criterion [13, Corollary 4.6], since then $(P_\theta)_{vv} = m_\theta/n > 0$, where m_θ is the multiplicity. We do not claim this case as new.

Questions. The problem of [2] remains open for trees in general, beginning with the trees of diameter 5 and 6, and so does the question whether the number $a(n)$ of Remark 8.1 is bounded.

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Tool and computational resource disclosure

This work was carried out on 26 and 27 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. Claude Code agents selected the problem, found the theorem and its proof, wrote the programs and the Lean code, carried out the literature searches described in the introduction, and drafted this text. Throughout, “we” refers to this workflow; the reading, searching and computing described in the paper were done by the agents. An adversarial checking agent (Claude Code, the same model), launched by the agent that found the proof, re-derived Lemmas 4.2, 4.3 and 5.4 and Proposition 5.3 and repeated the finite search; it found one sign error in a displayed equation, which did not affect the count and has been corrected. Its programs were not kept. An independent referee agent (Claude Code, the same model), which had not produced the work, then refereed all proofs and the correspondence between the Lean statements and the text, recomputed the finite search, the exceptional trees and further checks with its own programs (Section 6), and repeated the literature search. Its five minor corrections, concerning a cross-reference in the case $b^* \geq 13$, the description of the Lean coverage, a shorter proof of Proposition 5.1(c), the sign $-\theta$ in Proposition 5.1(e) and the description of the independent computations, have been applied. A second independent agent (Claude Code, the same model), which had produced neither the work nor the referee’s report, then reviewed this paper, re-derived its proofs and recomputed the computations of the proof with its own programs; its minor corrections, concerning the description of the Lean verification and some wordings, have been applied. The programs, logs and Lean code cited in this paper are in version 1.7.0 of the repository [4] (directories `certificates/switching/`, `lean/Research/` and `logs/`). The author is not a professional mathematician. The author has read the paper. The correctness of Theorem 1 rests on the written proofs and on the finite computation of Proposition 5.5, on which three independently written programs agree; the Lean formalisation covers only the steps listed in Section 7. AI systems are not authors of this paper; the author takes full responsibility for its content.

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